Seminars 9 & 10: IIR Filters

Questions and Final Answers

1. Recursive difference equations and initial rest

Consider the first-order recursive system

$$y[n] = 0.8y[n-1] + x[n]$$

with input $x[n] = u[n]$ and initial-rest conditions.

(a) State explicitly what initial-rest conditions mean for $x[n]$ and $y[n]$.

Answer. x[n]=0 and y[n]=0 for n<0; therefore y[−1]=0.

(b) Compute $y[0], y[1], y[2], y[3]$ directly from the recursion.

Answer.

$$y[0] = 1, \qquad y[1] = 1.8, \qquad y[2] = 2.44, \qquad y[3] = 2.952$$

(c) Write $y[n]$ for general $n \geq 0$ as a finite geometric sum.

Answer.

$$y[n] = \Sigma\_(k = 0)\^n\ (0.8)\^k$$

(d) Use the geometric-series formula to obtain a closed-form expression for $y[n]$.

Answer.

$$y[n] = 5(1 - 0.8\^(n+1)), \qquad n \geq 0$$

(e) Determine the final value approached by the step response and explain why it is finite.

Answer.

$$\lim\_(n \to \infty)\ y[n] = 5$$

2. Impulse response of a first-order IIR filter

Consider

$$y[n] = 0.6y[n-1] + 2x[n]$$

(a) Apply $x[n] = \delta[n]$ and calculate the first five samples of the impulse response $h[n]$.

Answer.

$$h[0{:}\,4] = \{2{,}1.2{,}0.72{,}0.432{,}0.2592\}$$

(b) Derive a closed-form expression for $h[n]$ valid for all n.

Answer.

$$h[n] = 2(0.6)\^n\ u[n]$$

(c) Explain why this is an infinite impulse response even though the input impulse has duration one sample.

Answer. Feedback keeps generating nonzero samples: h[n]=2(0.6)^n u[n] has infinite duration.

(d) Is the impulse response absolutely summable? What does your answer imply about stability?

Answer.

$$\Sigma|h[n]| = 5 < \infty \ \Rightarrow\ \text{stable}$$

(e) Repeat qualitatively for a recursive coefficient of 1.05 and compare the two responses.

Answer.

$$h[n] = 2(1.05)\^n\ u[n];\ \ \text{unstable}$$

3. System function from a recursive difference equation

Consider the first-order IIR filter

$$y[n] = 0.7y[n-1] + x[n] - 0.4x[n-1]$$

(a) Take the z-transform under initial-rest conditions and derive $H(z) = Y(z)/X(z)$.

Answer.

$$H(z) = (1 - 0.4z^{-1})/(1 - 0.7z^{-1})$$

(b) Identify the numerator coefficients b_0, b_1 and denominator coefficients.

Answer.

$$b = [1, -0.4], \qquad a = [1, -0.7]$$

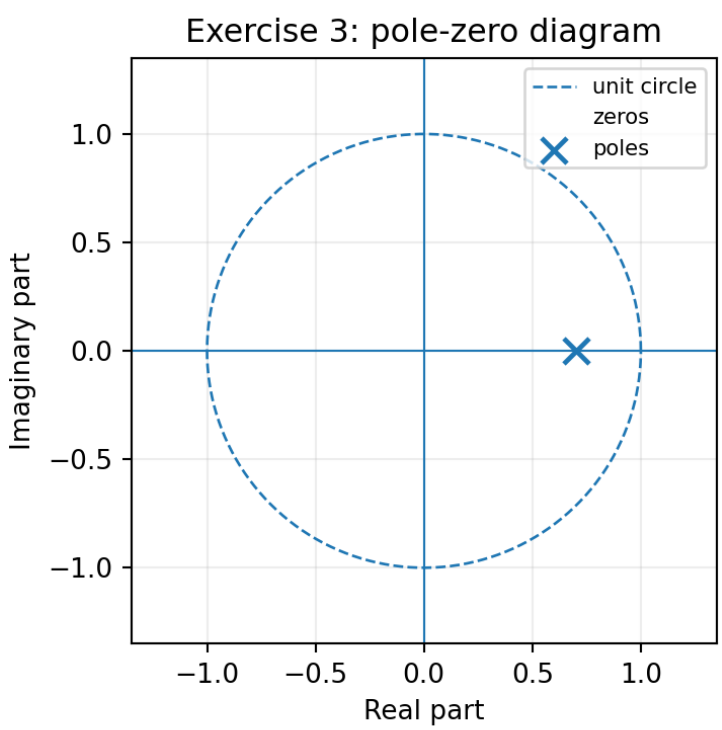
(c) Determine the pole and zero locations.

Answer.

$$\text{zero } z = 0.4; \;\; \text{pole } z = 0.7$$

(d) Sketch the pole-zero diagram and the unit circle.

Answer.



Pole-zero diagram.

(f) Explain which part of the structure is feed-forward and which part is feedback.

Answer. Feed-forward: x[n]−0.4x[n−1]. Feedback: +0.7y[n−1].

4. Region of convergence of an exponential sequence

Consider the causal sequence

$$x[n] = (0.75)^n u[n]$$

(a) Starting from the z-transform definition, show that the transform is a geometric series.

Answer.

$$X(z) = \Sigma\_(n = 0)\hat{}\infty\ (0.75z^{-1})\hat{}n$$

(b) Find $X(z)$ in closed form.

Answer.

$$X(z) = 1/(1 - 0.75z^{-1})$$

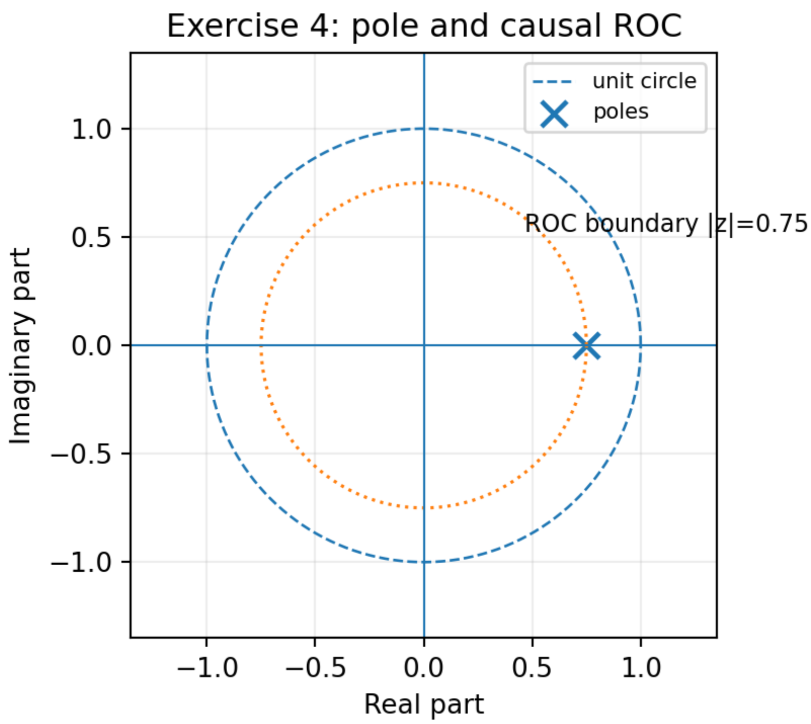
(c) Determine the region of convergence (ROC).

Answer.

$$\text{ROC: } |z| > 0.75$$

(d) Locate the pole in the z-plane and indicate the ROC on the same sketch.

Answer.



Pole and causal ROC.

(e) Does the ROC include the unit circle? Explain the significance of this for the existence of the frequency response.

Answer. Yes. The unit circle is inside the ROC, so the frequency response/DTFT exists.

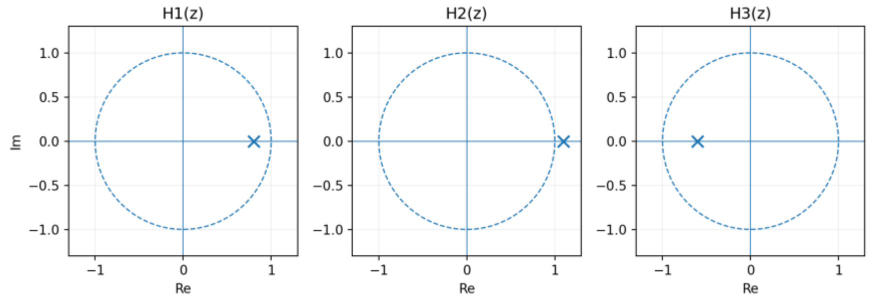
5. Poles, zeros, and stability

For each of the following causal system functions, find the poles and zeros and determine whether the system is stable:

$$H_1(z) = 1/(1 - 0.8z^{-1})$$
$$H_2(z) = (1 + z^{-1})/(1 - 1.1z^{-1})$$
$$H_3(z) = (1 - 0.5z^{-1})/(1 + 0.6z^{-1})$$

(a) Plot all pole-zero locations in the z-plane.

Answer.



Pole-zero locations.

(b) For each system, state the causal ROC.

Answer.

$$\mathrm{ROC}_1\text{: } |z| > 0.8;\;\; \mathrm{ROC}_2\text{: } |z| > 1.1;\;\; \mathrm{ROC}_3\text{: } |z| > 0.6$$

(c) Apply the pole-location stability criterion.

Answer.

$$H_1 \text{ stable};\;\; H_2 \text{ unstable};\;\; H_3 \text{ stable}$$

(d) Describe the expected long-time behaviour of each impulse response.

Answer. H_1: decaying; H_2: growing; H_3: decaying with alternating sign.

(e) Explain why a pole close to the unit circle produces a long transient.

Answer. Because |p|^n decays slowly when |p| is close to 1.

6. Frequency response of a first-order IIR filter

Consider

$$H(z) = (1 + z^{-1})/(1 - 0.8z^{-1})$$

(a) Determine the pole and zero locations and verify stability.

Answer.

$$\text{zero } z = -1;\;\; \text{pole } z = 0.8;\;\; \text{stable}$$

(b) Obtain $H(e^{j}\hat{\omega})$ by substituting $z = e^{j}\hat{\omega}$.

Answer.

$$H(e\text{^}(j\hat{\omega})) = (1 + e\text{^}(-j\hat{\omega}))/(1 - 0.8e\text{^}(-j\hat{\omega}))$$

(c) Evaluate the magnitude at $\hat{\omega} = 0, \pi/2, \pi$.

Answer.

$$|H(0)| = 10;\;\; |H(\pi/2)| \approx 1.1043;\;\; |H(\pi)| = 0$$

(d) Evaluate or sketch the phase at the same frequencies.

Answer.

$$\text{phase: } 0, \qquad -1.460 \text{ rad}, \qquad \text{undefined at } \pi$$

(e) Explain why the response is large near DC and zero at the Nyquist frequency.

Answer. Pole near +1 boosts low frequencies; zero at −1 cancels $\hat{\omega}=\pi$.

(f) For input $x[n] = \cos(\pi n/2)$ that exists for all n, write the steady-state output.

Answer.

$$y[n] \approx 1.1043 \cos(\pi n/2 - 1.460)$$

7. From a rational system function to the impulse response

Consider the causal system

$$H(z) = (1 - 0.2z^{-1})/(1 - 0.6z^{-1})$$

(a) Rewrite $H(z)$ as a direct term plus a first-order rational term.

Answer.

$$H(z) = 1 + (0.4z^{-1})/(1 - 0.6z^{-1})$$

(b) Use a known z-transform pair to obtain $h[n]$.

Answer.

$$h[n] = \delta[n] + 0.4(0.6)\hat{}(n - 1)u[n - 1]$$

(c) Verify your result by computing $h[0], h[1], h[2]$ from the difference equation.

Answer.

$$h[0] = 1, h[1] = 0.4, h[2] = 0.24$$

(d) Explain how the numerator changes the first part of the impulse response while the pole controls its exponential tail.

Answer. The numerator shapes the initial samples; the pole at 0.6 controls the exponential tail.

8. Inverse z-transform by partial fractions

Suppose

$$Y(z) = 1/((1 - 0.5z^{-1})(1 - 0.8z^{-1}))$$

(a) Find the two poles.

Answer.

$$\text{poles: } 0.5 \text{ and } 0.8$$

(b) Perform a partial-fraction expansion of $Y(z)$.

Answer.

$$Y(z) = -(5/3)/(1 - 0.5z^{-1}) + (8/3)/(1 - 0.8z^{-1})$$

(c) Assuming a causal sequence, compute the inverse z-transform $y[n]$.

Answer.

$$y[n] = [-(5/3)(0.5)\hat{}n + (8/3)(0.8)\hat{}n]u[n]$$

(d) Calculate the first four samples from your closed-form result.

Answer.

$$\{1, 1.3, 1.29, 1.157\}$$

(e) Which pole dominates the response for large n and why?

Answer. The pole at 0.8, because it is closer to the unit circle and decays more slowly.

9. Transient and steady-state response to a switched-on sinusoid

Consider the stable first-order filter

$$y[n] = 0.85y[n - 1] + x[n]$$

with $x[n] = \cos(0.3\pi n)u[n]$.

(a) Explain why the response contains both a pole-dependent transient and a sinusoidal steady-state component.

Answer. Response = pole-dependent transient + forced sinusoidal steady state.

(b) Determine $H(e^{j}0.3\pi)$ and hence the steady-state sinusoidal response.

Answer.

$$H \approx 1.17585e\text{^}(-j0.94175);\;\; y\_ss[n] \approx 1.17585\cos(0.3\pi n - 0.94175)$$

(c) State the general form of the transient component.

Answer.

$$y\_tr[n] = C(0.85)\text{^}n\, u[n]$$

(d) Does the transient decay or grow? Explain from the pole location.

Answer. Decays; the pole magnitude is 0.85<1.

(e) Approximately how many samples are needed for the transient amplitude to fall below 10% of its initial value?

Answer.

$$n \geq 15 \text{ samples}$$

10. Second-order poles and the difference equation

A second-order system has poles

$$p_{1,2} = 0.9e\text{^}(\pm j\pi/4)$$

(a) Form the denominator polynomial in z^{-1} with real coefficients.

Answer.

$$A(z) = 1 - 1.272792z^{-1} + 0.81z^{-2}$$

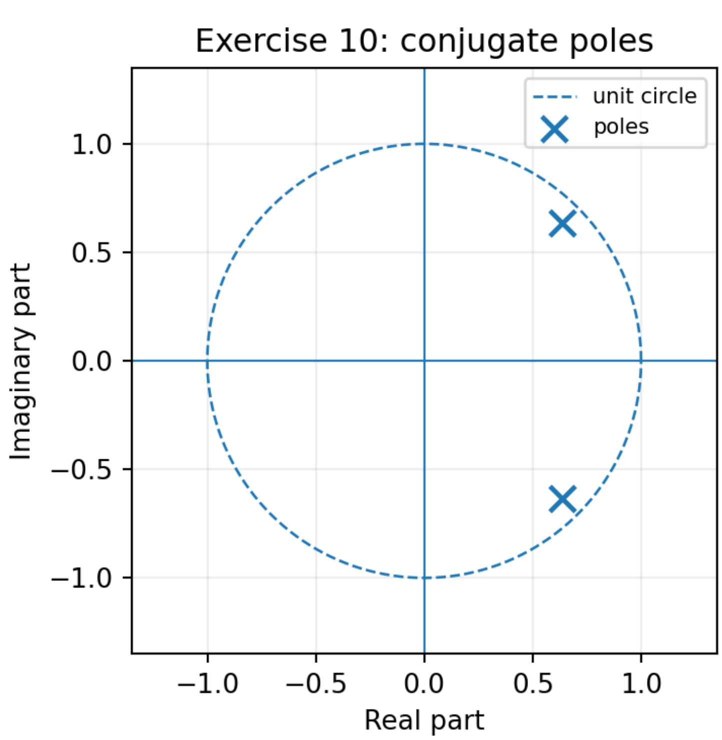
(b) Write a second-order recursive difference equation with numerator $B(z) = 1$.

Answer.

$$y[n] = 1.272792y[n-1] - 0.81y[n-2] + x[n]$$

(c) Plot the two poles and the unit circle.

Answer.



Pole locations.

(d) Is the causal system stable?

Answer. Stable: both poles have radius 0.9<1.

(e) State the oscillation frequency and explain the role of the pole radius in the transient envelope.

Answer. Oscillation frequency π/4 rad/sample; envelope proportional to 0.9^n.

11. Impulse response of a second-order resonator

Consider

$$H(z) = 1/(1 - 1.6\cos(0.2\pi)z^{-1} + 0.64z^{-2})$$

(a) Factor the denominator and identify the two poles.

Answer.

$$p_{1,2} = 0.8e\text{^}(\pm j0.2\pi)$$

(b) Determine their radius and angle.

Answer.

$$r = 0.8, \qquad \theta = \pm 0.2\pi$$

(c) Predict the qualitative shape of the causal impulse response before calculating it.

Answer. A stable damped sinusoid with envelope 0.8^n and frequency 0.2π.

(d) Use partial fractions or the standard conjugate-pole form to express $h[n]$ as a damped sinusoid.

Answer.

$$h[n] = (0.8)\text{^}n \sin((n+1)0.2\pi)/\sin(0.2\pi)\ u[n]$$

(e) What would change if the pole radius were moved from 0.8 to 1.02 without changing the angle?

Answer. Same oscillation frequency, but growing envelope 1.02^n; unstable.

13. Connecting the n, z, and frequency domains

For the causal system

$$H(z) = (1 - z^{-1})/(1 - 0.9z^{-1})$$

(a) Write the difference equation.

Answer.

$$y[n] = 0.9y[n-1] + x[n] - x[n-1]$$

(b) Determine the impulse response.

Answer.

$$h[n] = \delta[n] - 0.1(0.9)\text{^}(n-1)u[n-1]$$

(c) Find the pole and zero and discuss stability.

Answer.

$$\text{zero } z = 1;\ \text{pole } z = 0.9;\ \text{stable}$$

(d) Evaluate the frequency response at $\hat{\omega} = 0$ and $\hat{\omega} = \pi$.

Answer.

$$H(0) = 0;\ \ H(\pi) \approx 1.05263$$

(e) Explain how the zero at DC and the pole near $z = 1$ together shape the frequency response.

Answer. Zero at z=1 cancels DC; nearby pole at 0.9 makes the transition near DC sharp.

(f) Summarize how the same filter is represented in the time domain, z domain, and frequency domain.

Answer. Time: difference equation/h[n]. z: H(z) with pole/zero. Frequency: H(e^{jω̂}) on the unit circle.

14. Designing an IIR oscillator from desired poles

You want the impulse response of a second-order recursive system to contain a damped oscillation at normalized frequency $\hat{\omega}_0 = 0.25\pi$ with an envelope that decreases by a factor of 0.95 per sample.

(a) Choose the required complex-conjugate pole pair.

Answer.

$$p_{1,2} = 0.95e\text{^}(\pm j0.25\pi)$$

(b) Derive the real denominator polynomial.

Answer.

$$A(z) = 1 - 1.343503z^{-1} + 0.9025z^{-2}$$

(c) Write a second-order difference equation with $B(z) = 1$.

Answer.

$$y[n] = 1.343503y[n-1] - 0.9025y[n-2] + x[n]$$

(d) Is the system stable?

Answer. Stable.

(e) How would the sound or waveform change if the radius were increased to 0.995 while the pole angle stayed fixed?

Answer. Same frequency; much slower decay and longer ringing.

(f) How would it change if the angle were changed to 0.4π while the radius stayed 0.95?

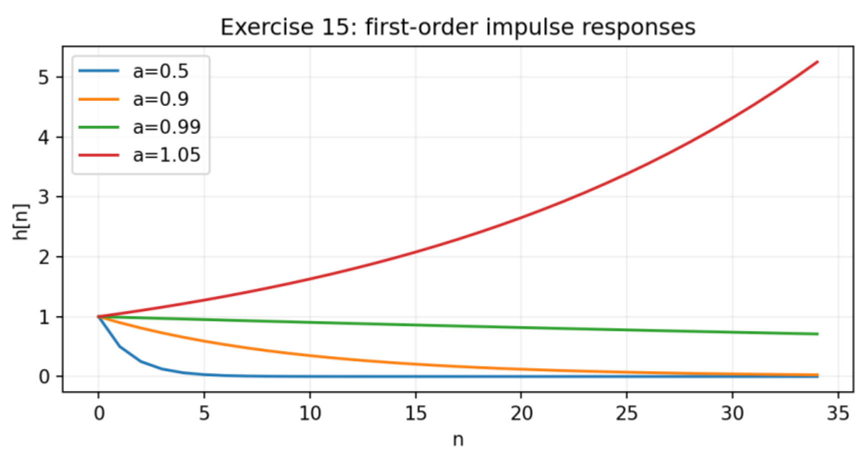
Answer. Frequency becomes 0.4π rad/sample; decay rate remains 0.95^n.

15. Computational investigation with Python

Use NumPy, Matplotlib, and scipy. signal to investigate IIR filters.

(a) For first-order systems $H(z) = 1/(1 - az^{-1})$ with $a = 0.5, 0.9, 0.99, 1.05$, plot the impulse responses and pole locations.

Answer.



Requested impulse-response comparison.

(b) For each case, classify stability and relate the pole radius to the decay or growth rate.

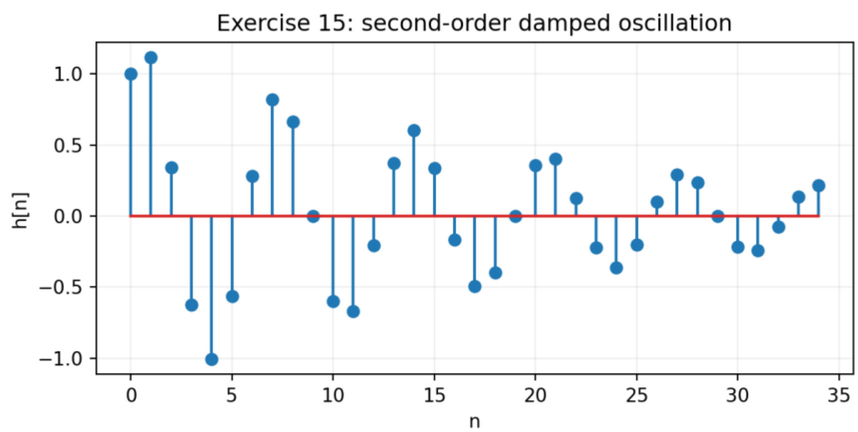
Answer. Stable: 0.5, 0.9, 0.99. Unstable: 1.05. Larger |a| gives slower decay.

(c) Use scipy.signal.freqz to plot the magnitude and phase of a stable first-order system and verify the response by filtering a sinusoid.

Answer. Use scipy.signal.freqz([1],[1,−a]) and verify that the steady-state sinusoid is scaled and phase-shifted by H(e^{jω̂₀}).

(d) For a second-order system with poles $0.95e\text{^}(\pm j0.3\pi)$, plot the impulse response and confirm that the oscillation frequency agrees with the pole angle.

Answer. Damped sinusoid at 0.3π rad/sample with envelope 0.95^n.



Second-order response.

(e) Use scipy.signal.lfilter to compare direct recursion with the output predicted by convolution with a long truncated impulse response.

Answer. Long truncated convolution approaches lfilter for a stable IIR; remaining error is the omitted impulse-response tail.

(f) Summarize how recursion, poles, stability, transients, and frequency response are related.

Answer. Recursion → poles → transient decay/growth and stability; pole angles → oscillation; H(e^{jω̂}) → steady-state frequency response.

Coverage map

Problems	Main concepts
1–3	Recursive difference equations, initial rest, impulse/step response, system functions, structures
4–6	Region of convergence, poles and zeros, stability, frequency response
7–9	Inverse z-transform, partial fractions, transient and steady-state behaviour
10–12	Second-order IIR filters, conjugate poles, damped oscillations, direct forms
13–14	Connections among domains and pole-based filter design
15	Computational verification of recursion, stability, frequency response, and pole behaviour